Ting Chong Xian

Singapore

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Junior Backend Engineer | Backend Intern

CS50 Graduate · GitHub App Developer · Production-Focused Backend Systems

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Projects

PR Brief — GitHub App for Automated Pull Request Summaries

Backend Engineer | Python, FastAPI, PostgreSQL, GitHub API, Stripe, Render

- Designed and built a **production-ready GitHub App** that automatically generates structured pull request summaries and posts them as PR comments and GitHub Check Runs.
- Implemented **event-driven webhook processing** for pull request lifecycle events (open, edit, synchronize) with idempotency safeguards to prevent duplicate processing.
- Integrated **FastAPI** with async request handling, background task execution, and structured logging for reliability and observability.
- Designed a **credit-based billing system** backed by PostgreSQL, including atomic credit reservation, refunds on failure, and Stripe Checkout integration.
- Implemented **security-first processing**, including GitHub webhook signature verification, heuristic secret detection prior to AI processing, and minimal-permission GitHub App scopes.
- Built a **fault-tolerant OpenAI request pipeline** with rate-limit handling, exponential backoff, and usage tracking via Prometheus metrics.
- Deployed and operated the service on **Render**, using **Neon PostgreSQL** for persistence and supporting operation with or without database availability.

Driftline — Continuous Configuration Drift Detection & Risk Intelligence Engine

Backend Engineer | Python, FastAPI, PostgreSQL, Async Pipelines

- Designed and built a **semantic drift intelligence pipeline** to detect, assess, and explain configuration and state changes across production systems.
- Architected a **multi-stage event pipeline** (Source → Snapshot → Diff → DriftSignal → Incident → RiskAssessment → Explanation → UsageEvent) with clear separation of concerns and monetization boundaries.

- Implemented **semantic snapshotting and structured diffs**, storing normalized state rather than raw text to enable meaningful change detection and magnitude scoring.
- Designed **immutable drift signals** and an **incident lifecycle system** with deterministic fingerprinting, escalation-only risk handling, and automatic resolution when drift disappears.
- Built a **risk assessment engine** that decouples change magnitude from risk severity, enabling prioritization of small-but-dangerous versus large-but-safe changes.
- Implemented **plan-gated, versioned LLM explanations** as a paid capability, with auditability, regeneration on escalation, and prompt/version tracking.
- Integrated **usage-based billing instrumentation** via explicit UsageEvent emission, enabling metering, plan enforcement, and future monetization expansion.
- Designed the system around **org-level isolation**, ensuring clean commercial boundaries for billing, access control, and scalability.

Technical Experience

Information Technology Officer (Part-time)

Setsco Services Pte Ltd · Singapore
Feb 2024 – Apr 2024

- Maintained the **security integrity of company laptops and workstations**, including OS hardening and security patch deployment.
- Performed **device inventory management**, provisioning and dispatching systems to departments with consistent configuration standards.
- Diagnosed and resolved **software and hardware issues** across end-user systems, reducing downtime and improving operational reliability.
- Developed and contributed to **automation scripts** using **Shell and PowerShell** to streamline repetitive IT workflows and administrative tasks.
- Gained hands-on exposure to **enterprise IT tooling, troubleshooting practices, and operational security considerations**.

Education & Certifications

CS50: Introduction to Computer Science — Harvard University
CS50: Introduction to Artificial Intelligence
CS50: Introduction to Programming with Python
CS50: Introduction to Cybersecurity

GCE A Levels — Tampines Meridian Junior College
Mathematics, Chemistry, Economics, History

Technical Skills

Languages: Python, SQL, C, Shell, PowerShell, JavaScript (basic)
Backend & APIs: FastAPI, REST APIs, GitHub Webhooks, Async Programming
Databases: PostgreSQL (Neon)
DevOps & Infra: Render, Environment-based config, CI workflows
Security: Webhook signature verification, secret detection, access scoping
Tools: Git, GitHub Apps, Stripe API, Prometheus

Awards

Certificate of Achievement — Ministry of Education
Singapore Youth Festival Performance Awards
National Youth Achievement Award (Silver, Bronze)